

CREATING A NEW TILE

OVERVIEW

As of May 2026, there are no plans to add any new tiles to Clean World, but should that change, this doc gives an overview of how to add new tiles. To handle the creation of new tiles, I made a developer tool called the Tile Factory (Accessible under “Tools” > “Create Tile”). The Tile Factory tries to streamline the process of making new tiles, but you’ll have to go back and fix quite a few messes it leaves in its trail. My apologies that the process is still somewhat overcomplicated.

The “Step-By-Step” section gives you a run-down of everything you’ll have to do. The following sections provide you with more specific information about how to carry out each step. One section at the end provides you with information about how to delete a tile permanently.

STEP-BY-STEP

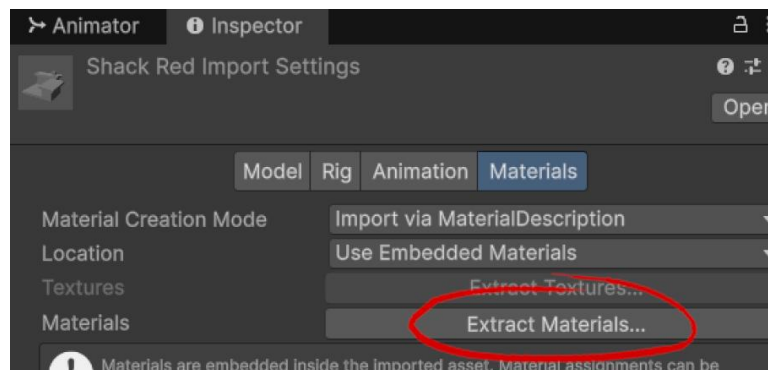
Here's the basic process for creating a new tile. Some of these steps are complicated & so are covered in more detail by later sections:

1. **Create the Model:** Create the 3D model for the tile in your program of choice (I personally use Blender), and export it as an “.fbx” file. Throw the file into the Unity project under the “Assets/Models” folder, preferably in its own new subfolder.
2. **Switch Model Materials To Dither Shader:** Extract the model's materials, then open the material editor from Unity's top menu bar (Tools > Set Object Material Shaders). In the window that the button opens, drag your fbx file into the “Model” box. Click “Set To Dither Shader”
 - a. See the section [“Setting Tile Material Shaders”](#) for more information.
3. **Create Tile:** Open the Tile Factory in Unity from the top Menu Bar (Tools > Create Tile), fill out all the tile's parameters, and click “Create New Tile”.
 - a. See the section [“Using the Tile Factory”](#)
4. **Adjust Model Scaling:** Open the new tile's prefab (which the Tile Factory created) at the location “Assets/Prefabs/Tiles/CurrentTiles”. Then, correct the 3D model's scaling and position. If the model isn't visible, you'll probably have to drag the model back into the prefab from its file.
5. **Reposition UI-Pop-Up Reference:** While still in the prefab, move the game object “PositionOfUIPopUps” to be slightly above the tile. It determines where the over-object UI Pop Ups appear.
6. **Assign Correct Script For Tile Type:** Ensure the correct kind of Tile script is assigned to the tile's prefab.
 - a. See the section [“Assigning the Correct Tile Script”](#)
7. **Set Up Activatable Tiles:** If the tile is an activatable tile, open the prefab and create the tile activation graphic, assigning it to the “Building Activated Graphic” parameter under the Tile script.
 - a. See the section [“Setting Up Activation Graphics”](#)
8. **Add Unlock Events:** In the Progression Manager, add a new progression event for when the tile should be unlocked.
 - a. See the doc [“Progression_System.pdf”](#). Specifically, the section [“Adding Unlock Events For New Tiles”](#)
9. **Test It:** Something usually goes wrong after finishing a new tile. Be sure to check that it works in all ways as expected.

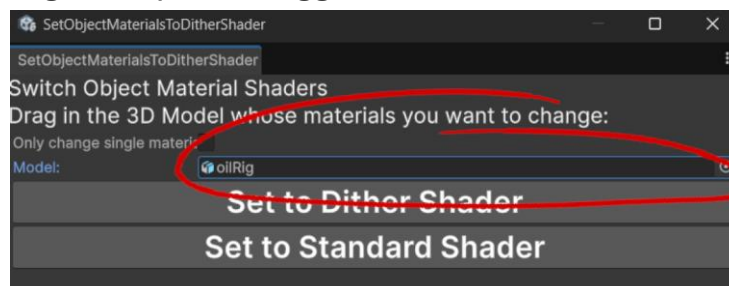
SETTING TILE MATERIAL SHADERS

Clean World has a feature that allows players to see through tiles when their mouse is hovering over the tile. If you want more information on that system, see the doc “Tile_Transparency_System.pdf”. In order for that system to work, every tile’s materials must be set to the Dither Shader. Here’s how to assign the dither shader to all the materials on a new tile:

1. **Extract Materials.** Select the tile’s model and in the inspector click “Extract Materials”



- a. If you don’t see the “Extract Materials” option, be sure that you have your “Location” parameter set to “Use Embedded Materials”.
2. **Set To Materials To Dither Shaders.** I made a tool to speed up the process of changing each material’s shader. You can open it from Unity’s top menu bar, at Tools > Set Object Material Shaders. In that window, there’s a slot labeled “Model”. Drag your model into it. In the following example, I dragged a model named “oilRig” into it.



3. In that window, click “Set To Dither Shader.”

USING THE TILE FACTORY

I apologize in advance for any janky-ness you may experience while using the Tile Factory. If it glitches out, I recommend closing the window and opening it again.

You can open the Tile Factory in Unity from the top menu, Tools > Create Tile. This window should appear:

TILE CREATOR 3000

Enter all of the data for your new tile. When you're done, press "Create New Tile"
If you need any help with this tool, or if you need anything, ask Graydon

FBX File: None (Game Object)

Button Image: None (Sprite) [Select]

Enter Tile Name: Untitled Tile

Money Per Year: 0

Pollution Per Year: 0

Housing Capacity: 0

Required Number of Emplo: 0

Build Cost: 0

Tile is a Residence: ☐

Tile can be Activated/Deac: ☐

Must Connect With Roads: ☐

Create New Tile

Important: Be sure you assign the FBX file and Button Image fields (even if you plan on changing them later) before creating a new tile. If you don't, the Tile Factory likes going haywire.

- **FBX file:** The 3D model for the tile. Must be FBX.
- **Button Image:** Sprite for the button that will place the tile. (Note: "Texture Type" must be "Sprite" and "Sprite Mode" must be "Single")
- **Enter Tile Name:** The name that will be put on all the components of the new tile.
- **Money per Year, Pollution Per Year, Housing Capacity** – Pretty self-explanatory.

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- **Required Number of Employees** – For workplaces; how many employees they require.
 - **Build Cost** – How much it costs to place the tile.
 - **Tile is a Residence** – True for tiles where people live.
 - **Tile Can Be Activated/Deactivated** – For tiles that only produce money/carbon when certain conditions are met (they need employees, must be connected by roads, etc). They are deactivated when conditions aren't met.
 - **Must Connect With Roads** – True for tiles that only work when they're connected by roads.

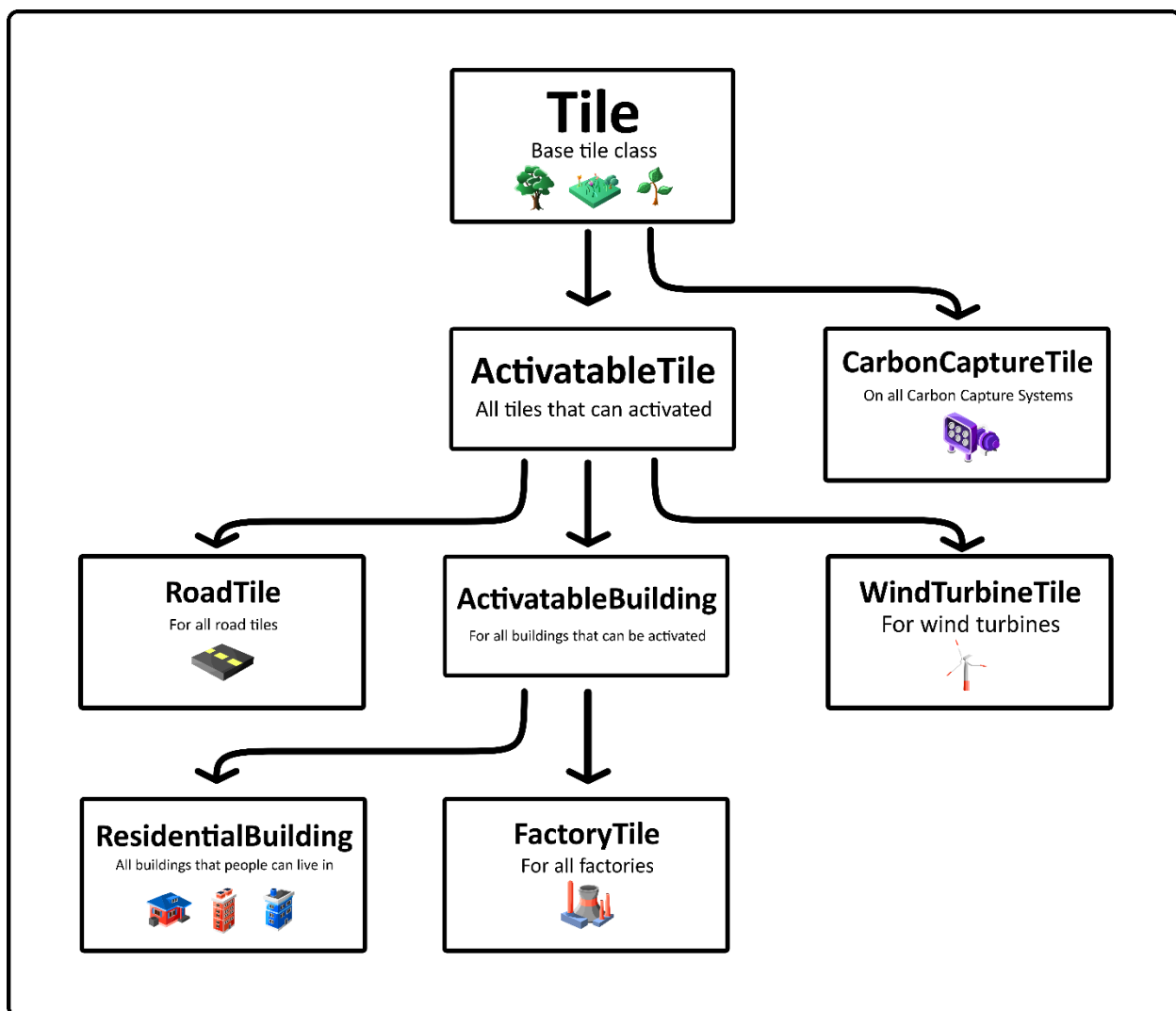
When you're done filling out all the fields, click "Create New Tile". If any errors occur, you may need to close the Tile Factory window and fill it out again, this time fixing the problems.

Once you press "Create New Tile", the Tile Factory will create a scriptable object for your new tile, a new prefab for it, a button for it, and do its best to connect everything correctly.

ASSIGNING THE CORRECT TILE SCRIPT

Every tile needs to have an instance of the Tile class on it, or a class that inherits from the Tile class. When you create a new tile, the Tile Factory tries to pick the best fit, but you'll still need to ensure that it's the right one for your tile. The primary ones you'll likely need to use are Tile, Activatable Tile, or Residential Building. Please note that this system is not a very good use of inheritance, and it caused me some unnecessary headaches down the road.

Here's a chart of how all the different kinds of tile scripts inherit from the Tile class:



SETTING UP ACTIVATION GRAPHICS

When tiles are activated, their activation graphic game object, which is a child of the tile, is turned on. The activation graphic game object can have particle effects for chimneys, yellow sprites for lit up windows, or basically anything else you want. You just want to make sure that it's clear to the player that the tile is activated when it's turned on.

Here's an example of the Mega Apartment's activation graphic game object, which is composed of several brightly colored sprites and two particle emitters:

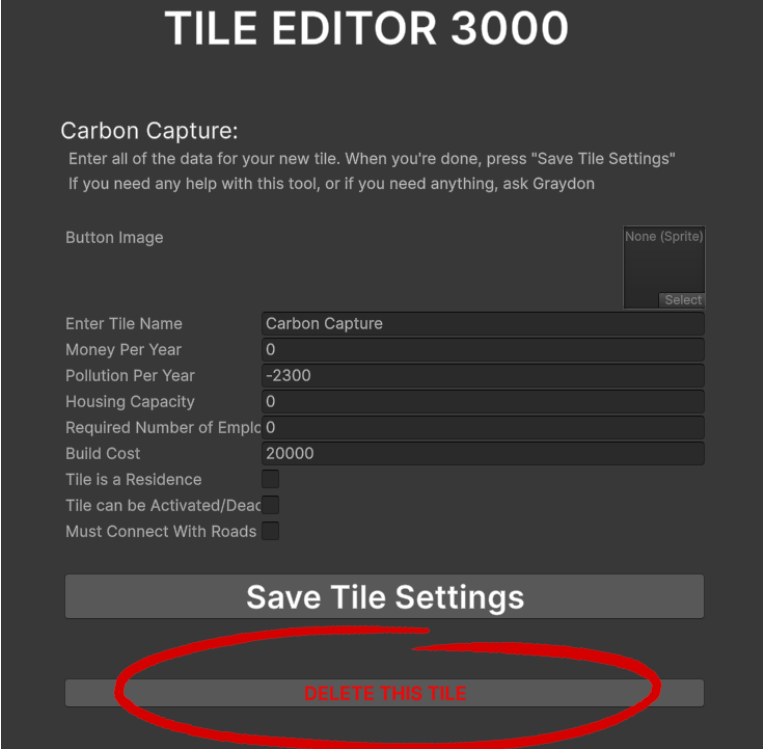


After completing the activation graphic game object, disable it so that it's not visible and in the inspector, assign it to the "Building Activated Graphic" parameter under the Tile script (Or Residence Tile/ Activatable Building script) of the new tile's prefab.

ALSO, DELETING A TILE

To delete a tile, you can use another developer tool called the Tile Editor. It allows you to edit a tile's properties. Additionally, it ensures that when a tile is deleted, every object and file associated with the tile is also deleted.

1. Open the Tile Editor in Unity from the top menu: Tools > Tile Editor.
2. It will open this window. Press "Delete This Tile" at the bottom:



TILE EDITOR 3000

Carbon Capture:
Enter all of the data for your new tile. When you're done, press "Save Tile Settings"
If you need any help with this tool, or if you need anything, ask Graydon

Button Image None (Sprite) Select

Enter Tile Name	Carbon Capture
Money Per Year	0
Pollution Per Year	-2300
Housing Capacity	0
Required Number of Emplo	0
Build Cost	20000
Tile is a Residence	☐
Tile can be Activated/Deac	☐
Must Connect With Roads	☐

Save Tile Settings

DELETE THIS TILE

3. In the second window that pops up when you click that, double check that you're deleting the right tile (the title of the window will have the tile's name). If so, then confirm that you want to delete the tile.

Clean World Documentation

Creating a New Tile



-Graydon Bruyn, May 2026